INCH POUND
MIL-PRF-3098/56F
27 August 1997
SUPERSEDING
MIL-C-3098/56E
10 April 1995

PERFORMANCE SPECIFICATION SHEET

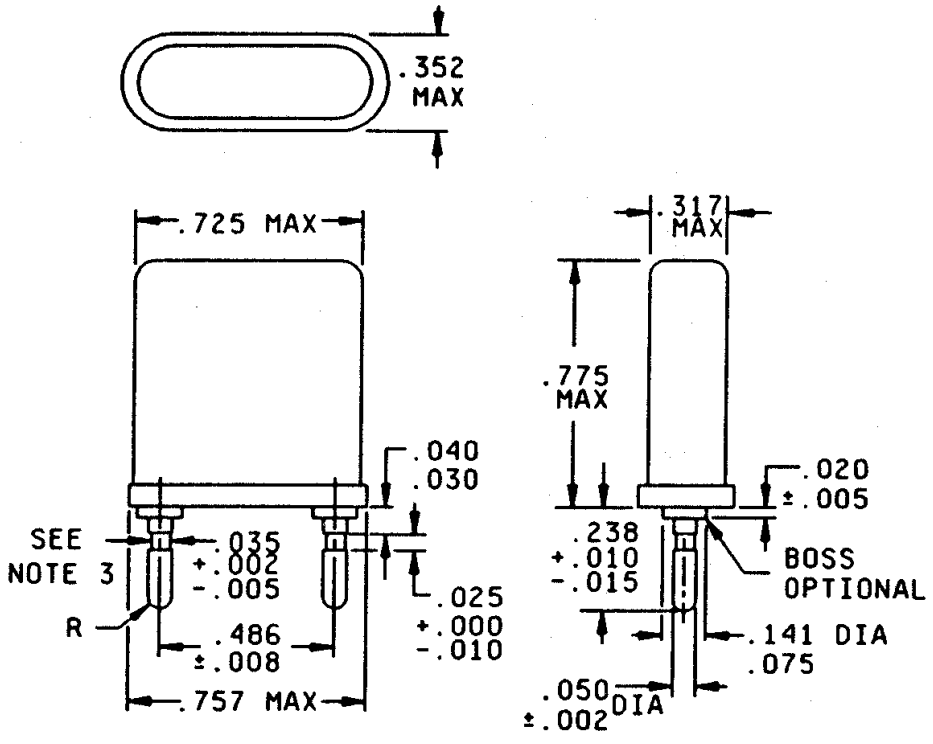
CRYSTAL UNIT, QUARTZ, CR85/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; series resonance.

Inches	mm
.002	.05
.005	.13
.008	.20
.010	.25
.015	.38
.020	.51
.025	.64
.030	.76
.035	.89
.040	1.02
.050	1.27
.075	1.91
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR85/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Frequency tolerance:

Primary operating temperature range: ±20 parts per million (ppm).

Secondary operating temperature range: ±30 ppm.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.85	520
0.85+ to 0.90	480
0.90+ to 1.00	440
1.00+ to 1.12	400
1.12+ to 1.25	380
1.25+ to 1.37	340
1.37+ to 1.50	300
1.50+ to 1.62	280
1.62+ to 1.75	250
1.75+ to 1.87	220
1.87+ to 2.00	190
2.00+ to 2.12	170
2.12+ to 2.25	150
2.25+ to 2.60	130
2.60+ to 3.00	90
3.00+ to 3.40	70
3.40+ to 3.75	52
3.75+ to 4.00	45
4.00+ to 5.00	37
5.00+ to 7.00	25
7.00+ to 10.00	20
10.00+ to 15.00	18
15.00+ to 20.00	15

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Mode of oscillation: Fundamental.

Operating temperature range:

Primary operating temperature range: -40°C to +90°C, inclusive.

Secondary operating temperature range: -55°C to -40°C, and +90°C to +105°C, inclusive.

Drive level: 1.0 mW maximum.

Resonance: Series.

Capacitance, shunt: 7.0 pF maximum.

Shock (specified pulse):	0.8 MHz to 2.0 MHz	2.0+ MHz to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent.	±10 percent

Vibration: Method 201 of MIL-STD-202.	0.8 MHz to 2.0 MHz	2.0+ MHz to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent.	±10 percent

Thermal shock:	0.8 MHz to 2.0 MHz	2.0 MHz to 20.0 MHz
Frequency change permitted:	±10 ppm.	±5 ppm
Equivalent resistance change permitted:	±15 percent.	±10 percent

Aging:

Frequency change permitted: ±5 ppm.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air force - 85

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - CC

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